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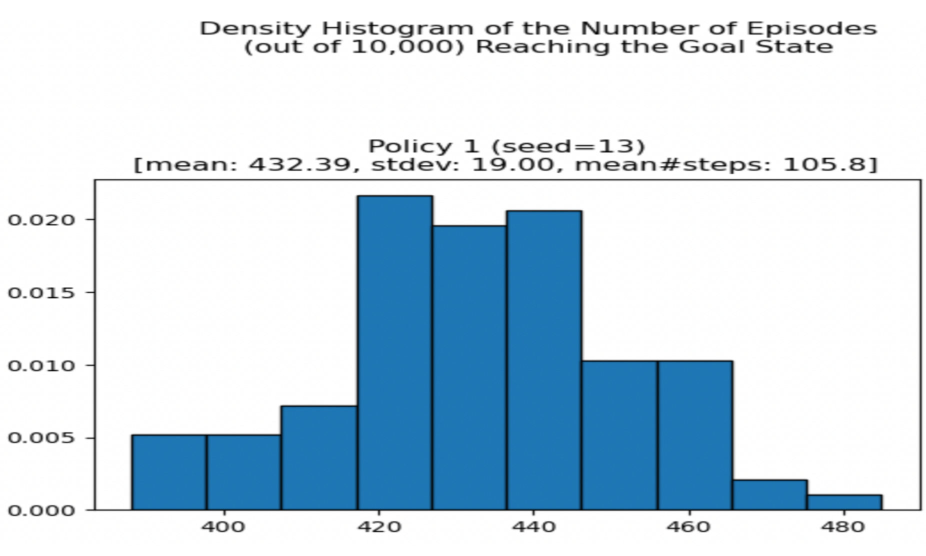
RESULTS

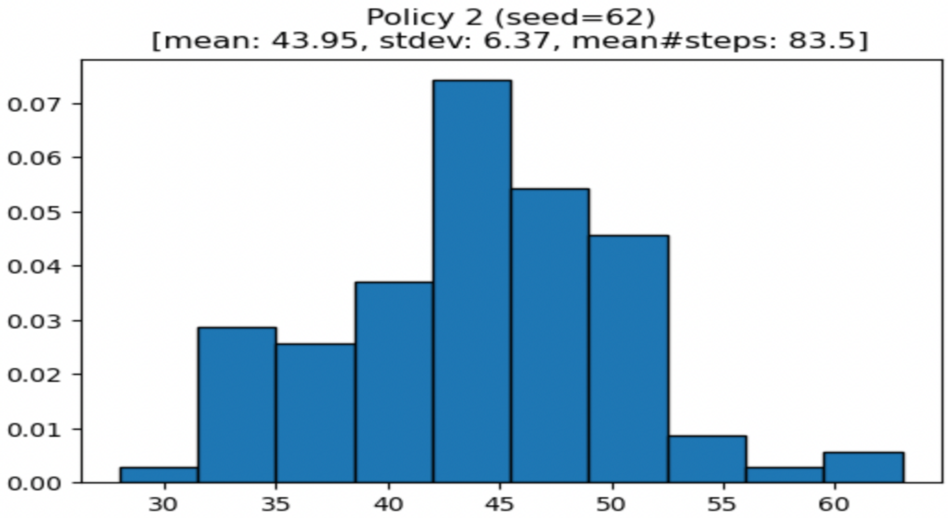
PART – 1: FIXED POLICIES

```
==== Top 2 Policies ====

1: Policy Seed 13
*** Policy ***
[[3 3 3 3 0 3 3 1]
 [0 0 3 3 2 2 3 0]
 [3 3 0 3 3 1 1 3]
 [3 0 2 1 1 3 2 1]
 [3 2 3 0 3 3 2 1]
 [1 0 1 3 1 3 1 2]
 [1 2 1 3 2 0 3 2]
 [2 3 1 0 2 0 0 2]]

2: Policy Seed 62
*** Policy ***
[[1 2 2 2 3 3 0 1]
 [0 2 2 3 0 2 3 3]
 [1 3 1 0 1 3 2 0]
 [1 0 1 1 1 0 3 1]
 [1 2 2 1 1 3 2 2]
 [1 3 2 1 1 2 2 0]
 [3 0 1 2 2 2 2 1]
 [3 0 1 2 3 3 3 3]]
```





Histograms were plotted for both the policies with seeds 13 and 62. X-axis represents number of goals reached and Y-axis represents frequency of occurrences.

Summary Comments:

Key Observations:

- We selected the top two policies out of 5 based on the highest average goals achieved.
- Policy Seed 13 performed significantly better than Policy Seed 62 in meeting the goal.
- The standard deviation of Policy 13 is slightly higher, indicating some variation in performance.

Performance Comparison:

Metric	Policy Seed 13	Policy Seed 62
Mean Goals	432.39	43.95
Standard Deviation	19.0	6.37
Mean Steps per Goal	105.8	83.5

Code steps:

- We have initialized FrozenLake-v1 environment with 8*8 map which is slippery
- We created 5 policies with different seeds running each policy for 100 times with 10,000 episodes
- Then selected the top 2 best policies based on mean goals reached, displayed the policy and the histogram

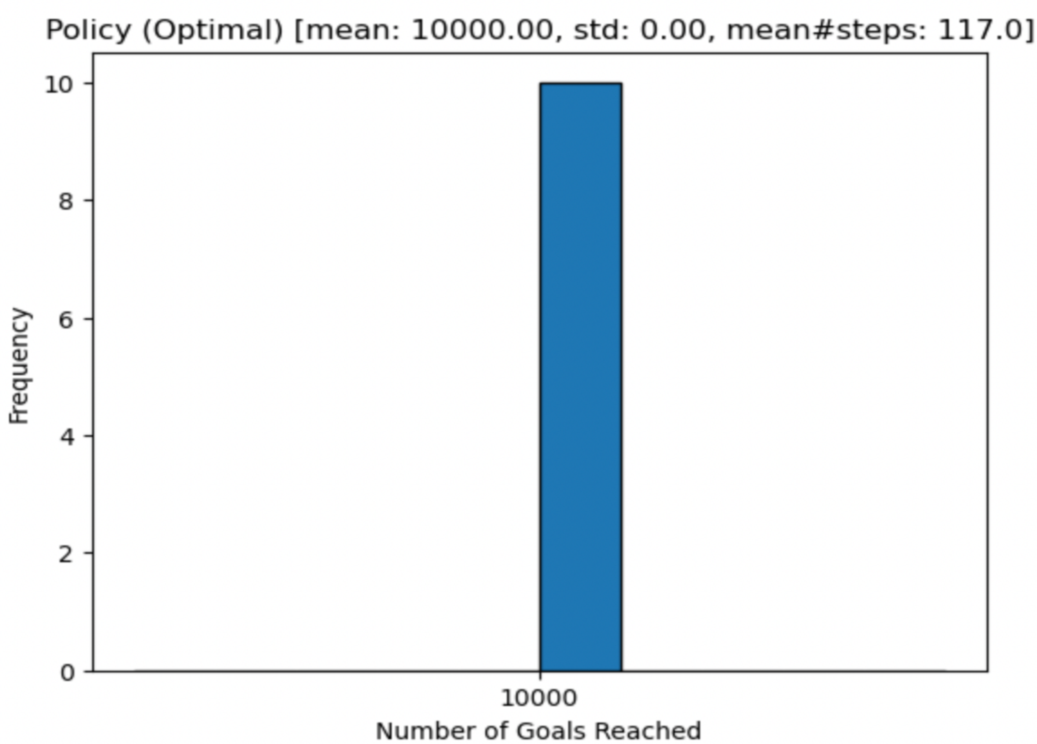
Insights:

A higher mean goal count does not always mean the policy is more efficient in reaching goals. Fewer steps per goal (lower mean steps) can indicate a more direct path but also means higher risk.

PART – 2: OPTIMAL POLICY BY VALUE ITERATION

```
*** Optimal Policy ***
[[3 2 2 2 2 2 2 2]
 [3 3 3 3 3 3 3 2]
 [0 0 0 0 2 3 3 2]
 [0 0 0 1 0 0 2 2]
 [0 3 0 0 2 1 3 2]
 [0 0 0 1 3 0 0 2]
 [0 0 1 0 0 0 0 2]
 [0 1 0 0 1 2 1 0]]

Final Values:
[[0.99823323 0.99831044 0.99842197 0.99854354 0.99866612 0.99878334
  0.99888814 0.99896483]
 [0.99821397 0.99827276 0.99836936 0.99848462 0.99860719 0.99873171
  0.99886003 0.99901008]
 [0.99683199 0.97525654 0.92389215 0.         0.85531987 0.94498778
  0.98096596 0.99909858]
 [0.99558378 0.93078946 0.79815834 0.47350677 0.62242089 0.
  0.94378534 0.99922649]
 [0.99451946 0.82168741 0.53991636 0.         0.53848733 0.61046658
  0.85120246 0.99938819]
 [0.99368164 0.         0.         0.16763079 0.38261111 0.44174102
  0.         0.99957664]
 [0.99310381 0.         0.19328935 0.1202984  0.         0.33216797
  0.         0.99978359]
 [0.99280901 0.72610605 0.45962631 0.         0.27738468 0.55477288
  0.77738468 0.        ]]
```



Histogram displays a single peak at 10,000, ensuring that the correct approach consistently reaches the target in each iteration. Since there is no variation in the results, the x-axis is set to highlight this single value.

Summary Comments:

Optimal Policy Characteristics:
The derived optimal policy ensures consistent goal completion in all cases.
The value (V) table shows that states closer to the goal have higher values, reinforcing optimal behavior.

Policy Evaluation:

Metric	Optimal Policy
Mean Goals	10,000
Standard Deviation	0.00
Mean Steps per Goal	117.0

Insights:
The best action that leads to the goal is assigned to the most state and the agent effectively avoids holes, proving that value iteration works. Histogram clearly shows a single peak meaning that policy always reaches the goal.

Do the Actions Make Sense?

Yes, it makes sense. The best way to win is not always taking the shortest path. Even if moving away from the goal seems like a bad idea the smart way is to think ahead and avoid holes. The policy looks at all possible future moves and picks the safest and most successful way to reach the goal. In this case the best choice is not the fastest one. Instead, the policy chooses a path that keeps the agent safe, reduces risk, and gives the best chance of reaching the goal successfully.

General Reflections:

- Talking about the assignment, it was a very interesting experience for me. I was able to understand reinforcement learning concepts. Value iteration and analyzing different policies helped me realize how different strategies can impact performance in a such difficult environments.

- The exercise seemed quite difficult because I had to ensure that my policy evaluation correctly computed the best actions. I struggled with handling state transitions and keeping track of value updates properly. I understood the concept of maintaining two value tables. One for the current values and one for updated values it became much easier to implement.

- One of the biggest difficulties I faced was ensuring that the histogram properly represented the results. Since my optimal policy always reached the goal, my histogram initially looked incorrect because all values were the same. By adjusting the x-axis range, helped make it more visually meaningful. Also, by interpreting the value function table, but once I saw how states closer to the goal had the higher values which made sense later.

- I would try experimenting with different values of gamma to see how discounting affects in long term rewards in case I had to do this assignment again. I would also test different starting policies to see how much they influence the learning process. I would spend more time for visualizing state transitions to understand the policy changes over iterations.